

Pfutsero Phek Road - Summary of Cost

Bill No.	Description	Amount (Rs.)	Figure in Crore
A	Civil Works		
1	SITE CLEARANCE AND DISMANTALING	1,53,18,649	1.53
2	EARTH WORKS	1,23,12,45,505	123.12
3	GRANULAR SUB-BASE AND BASE COURSES	1,31,77,33,700	131.77
4	BITUMINOUS COURSES AND CONCRETE PAVEMENTS	68,63,33,109	68.63
5	CROSS DRAINAGE WORKS	1,70,75,62,820	170.76
6	BRIDGES / UNDERPASS	6,16,38,010	6.16
7	REPAIR OF EXISTING STRUCTURES	0	0.00
8	DRAINAGE AND PROTECTIVE WORKS	3,21,18,65,097	321.19
9	TRAFFIC SIGNS, MARKINGS AND ROAD APPURTENANCES	17,47,90,136	17.48
10	MISCELLANEOUS	6,78,61,785	6.79
Sub Total for Civil Works (A)		8,47,43,48,810	847.43
B	Utility Shifting Cost	1,30,00,000	1.30
Sub Total (A+B)		8,48,73,48,810	848.73
C	Physical Contingencies @5% on (A+B)	42,43,67,441	42.44
Grand Total(A+B+C)		8,91,17,16,251	891.17
Length of the package Road (km)		62.6	Km
Cost per km on TPC		14.2	Crore

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

RTES Ltd.
Highway Division

**BILL OF QUANTITIES (BOQ)
BILL NO. 1:GENERAL AND SITE CLEARANCE**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
1.01	Clearing and Grubbing of Roadway as per Technical Specification Clause 201.	Ha	175	55,705	97,53,461
1.02	Dismantling of Existing Structures or any Existing Civil Works within Right of Way, as per Technical Specification Clause 202.				
	(i) Brick Masonry	cum	Nil	357	-
	(ii) Stone Masonry	cum	8,080	357	28,84,560
	(iii) Lime/Plain Cement Concrete in all grades	cum	700	542	3,79,400
	(iv) Reinforced / Pre-stressed Cement Concrete in all grades	cum	120	916	1,09,920
	(v) Bituminous courses	cum	1,966	336	6,60,724
	(vi) Non Bituminous courses	cum	Nil	578	-
	(vii) Cement concrete Pavements	cum	Nil	1,776	-
	(viii) Ordinary KM stone/Guard stone/Sign Post	No.	5	253	1,265
	(ix) Hectometre Stone	No.	Nil	51	-
	(x) 5th KM stone	No.	Nil	435	-
	(xi) Kerb stone	m	Nil	17	-
	(xii) All types of Fencing including posts, foundation concrete, back filling of pit etc. complete.	m	Nil	55	-
	(xiii) RCC Railing	m	Nil	258	-
	(xiv) Guard Rails/W Beam Crash Barrier	m	Nil	87	-
	(xv) Brick Pitching / Brick soling	cum	Nil	229	-
	(xvi) Removal of abandoned Telephone / Electric Poles and Lines including excavation and dismantling of foundation concrete and lines under the supervision of concerned department.	m	Nil	194	-
	(xvii) Removing all types of hume pipes and stacking serviceable material with all leads & lifts including earthwork and dismantling of masonry works.				
	(a) Up to 600 mm dia		Nil		-
	(b) Above 600 mm to 900 mm dia	m	Nil	206	-
	(c) Above 900 mm dia	m	2,220	279	6,19,380
	(xviii) Dismantling stone pitching/ dry stone spalls	m	Nil	478	-
	(xix) Dismantling of CI Water Pipe Line up to 1000 mm dia.	cum	Nil	245	-
		m	Nil	146	-
Carried to Page Collection					1,44,08,710

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

RTIES Ltd.
Highway Division

**BILL OF QUANTITIES (BOQ)
BILL NO.5: CROSS DRAINAGE STRUCTURES (CULVERTS)**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
5.01	Excavation for Foundation of Structure, as per drawings and Technical Specification, in (a) All types of soils and Ordinary Rocks as per Technical Specification Clause 304. (b) Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304. (c) Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304. (d) Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum cum cum cum	37,871 Nil Nil 88,365	253 416 409 546	95,81,363 - - 4,82,47,290
5.02	Plain Cement Concrete (M 15 Grade) for Levelling Course below Headwall, Abutment, Wing wall, Around foundation in rock, Approach slab, Chutes, Catch Pits etc. complete as per drawings and Technical Specification Section 1700, 2100, 2200, 2500 and 2700.	cum	3,423	12,744	4,36,22,712
5.03	RCC/PCC for open foundations complete excluding reinforcement as per drawings and Technical Specification Section 1700 and 2100 (a) M 25 Grade (b) M 30 Grade	cum cum	Nil 4,456	14,978 14,982	- 6,67,59,792
5.04	RCC / PCC for Substructure, Wing wall, Retaining wall, Box Cell Structures etc. excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2200. (a) M 25 Grade (b) M 30 Grade	cum cum	Nil 20,615	15,881 15,924	- 32,82,73,260
5.05	Coursed Rubble Masonry in 1:3 Cement Mortar for Foundation and Substructure complete as per Drawing and Technical Specification Clause 1400, 2100 and 2200.	cum	Nil	11,843	-
5.06	RCC (M 30 Grade) for Slab Culvert Superstructure excluding reinforcement complete as per drawings and Technical specifications Section 1700 and 2300.	cum	Nil	17,269	-
5.07	Steel Reinforcement (Fe 500D) for Foundations, Substructures and superstructures etc. complete as per drawings and Technical Specification Section 1600.	MT	2,905	91,126	26,47,21,030
5.08	Approach Slab (RCC M 30 Grade) Including reinforcement complete as per drawings and Technical Specification Section 2700.	cum	3,680	19,373	7,12,92,640
5.09	RCC Crash Barrier (M 40 Grade) excluding cost of reinforcement complete as per drawings and Technical Specification Section 1700 and 2700	cum	Nil	19,383	-
5.10	RCC (M 30 Grade) comprising cast in-situ Post and Precast Hand rails including reinforcement complete as per drawings and Technical Specification Section 1600, 1700 and 2700.	Rm	Nil	3,245	-
Carried to Page Collection					83,24,98,087

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

RITES Ltd.
Highway Division

**BILL OF QUANTITIES (BOQ)
BILL NO.5: CROSS DRAINAGE STRUCTURES (CULVERTS)**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
5.11	RCC Pipes, having diameter (in mm) as detailed below, for culverts complete as per drawings and Technical Specification Section 2900				
	(a) 900	m	Nil	18,750	-
	(b) 1000	m	Nil	20,833	-
	(c) 1200	m	4,533	23,991	10,87,51,203
	(d) 1500	m	Nil	29,989	-
5.12	Filter Media behind Earth Retaining Structures (abutment, splayed wing wall, retaining wall, return wall etc.) complete as per Technical Specification Section 2500	cum	12,504	3,845	4,80,77,880
5.13	Back filling behind Earth Retaining Structures (abutments, splayed wing walls, return wall, box cell structures etc.) Complete as per drawings and Technical Specification Clause 305.	cum	62,005	3,980	24,67,79,900
5.14	Tar Paper Bearing complete as per drawings and Technical Specification Section 2000.	sqm	Nil	140	-
5.15	Expansion Joint (Filler Joint) complete as per Drawings and Technical Specification Section 2600.	m	Nil	372	-
5.16	Expansion Joint (Asphaltic Plug Joint Type) Complete as per Drawings and Technical Specification Section 2600	m	3,004	1,777	53,38,108
5.17	Drainage spouts complete as per drawings and Technical Specifications Section 2700.	nos	544	4,374	23,79,456
5.18	Filter Material underneath Boulder / Concrete Block Pitching complete as per drawing and Technical Specification Section 2500.	cum	Nil	4,049	-
5.19	Boulder / Concrete Block Pitching on Embankment Slopes, In front of Abutment and In Launching Apron complete as per drawings and Technical Specification Section 2500.	cum	Nil	4,283	-
5.20	Boulder / Concrete Block Pitching In hand woven Wire Crates on Embankment Slopes in front of Abutment and in launching apron complete as per drawings and Technical Specification Section 2500	cum	21,044	6,311	13,28,08,684
5.21	Mechanically Woven Revetment / Gabion mattress filled with Boulder / Concrete block for erosion control, river training and protection works complete as per drawings and Technical Specification Section 2500	cum	Nil	6,325	-
5.22	Plain Cement Concrete (M 15 Grade) for Cement Concrete Bedding for Protection works complete as per drawings and Technical Specification Section 1700 and 2500.	cum	4,117	13,254	5,45,66,718
5.23	Flat Stone Apron for Bed Protection Works complete as per drawings and Technical Specification Section 1300, 1400 and 2500	cum	Nil	11,217	-
5.24	Lettering and Number of Culverts as per drawings and Additional Technical Specification A-3.	nos	21,800	1	15,260
5.25	Painting with water based cement paint on concrete surface as per additional Technical	sqm	2,235	196	4,38,060
5.26	Weepholes (AC/PVC/HDPE) in earth retaining structure, complete as per drawings and Technical	nos.	17,440	658	1,14,75,520
5.27	RCC/PCC for rigid flooring, buffer pier, blocks chutes etc. excluding reinforcement complete as per	cum	16,606	15,924	26,44,33,944
Carried to Page Collection					87,50,64,733

ABSTRACT OF QUANTITIES
BILL NO. 6: BRIDGES

Item	Description	Unit	No.	Length (m)	Breadth (m)	Height / Depth(m)	Quantity
6.11	Elastomeric Stopper Pad for Seismic Restrainer as per drawings and Technical Specification 2000	cu.cm					-
6.12	Expansion Joints, of following Type as per drawings and Technical Specification Section 2600						
a	Filler	m	As per annexure 6.1				52
b	Asphaltic Plug	m					-
c	Strip Seal Type capable of taking						
i	40 mm movement	m					-
ii	80 mm movement	m					-
iii	160 mm movement	m					-
6.13	Helical Void Former for Voided Concrete Slabs including 20 mm diameter tie rod and seating joints as per drawings and Technical Specification Section 2300 and Additional Technical Specification A – 4.	m					-
6.14	Drainage Spouts as per Technical Specification Section 2700	nos	As per annexure 6.1				25
6.15	Approach Slab (RCC, M 30 Grade) including reinforcement as per drawings and Technical Specification Section 2700	cum	As per annexure 6.1				66
6.16	RCC (M 30 Grade) comprising cast in-situ Post and Precast Hand rails including reinforcement complete as per drawings and Technical Specification Section 1600, 1700 and 2700.	Rm	As per annexure 6.1				64
6.17	RCC Crash Barrier (M 40 Grade) excluding cost of reinforcement complete as per drawings and Technical Specification Section 1700 and 2700	cum	As per annexure 6.1				15
6.18	Filling of Selected imported granular materials below structure as per drawings and Technical Specification Clause 305, 1000 and 2900	Cum					-
6.19	Weep holes (AC/PVC/HDPE) in Earth Retaining Structures (Abutments, Splayed Wing wall, Median Wall, Retaining Wall etc.) complete as per drawings and Technical Specification Section 2700.	nos	As per annexure 6.1				546
6.20	Back filling behind Earth Retaining Structures (abutments, splayed wing walls, return wall, box cell structures etc.) Complete as per drawings and Technical Specification Clause 305.	cum	As per annexure 6.1				3,307
6.21	Filter Media behind Earth Retaining Structures (abutment, splayed wing wall, retaining wall, return wall etc.) complete as per Technical Specification Section 2500	cum	As per annexure 6.1				351
6.22	Boulder / Concrete Block Pitching on Embankment Slopes, In front of Abutment and In Launching Apron complete as per drawings and Technical Specification Section 2500.	cum					-
6.23	Boulder / Concrete Block Pitching in hand woven Wire Crates on Embankment Slopes in front of Abutment and in launching apron complete as per drawings and Technical Specification Section 2500.	cum					-

ABSTRACT OF QUANTITIES
BILL NO. 8: DRAINAGE AND PROTECTIVE WORKS

Item No.	Description	Unit	No	Length (m)	Breadth (m)	Height / Depth(m)	Quantity
8.05	Gabion Structure for Retaining Earth with segments of mechanically woven (double twist) wire crates filled with boulders including non-woven geotextile as filter media complete as per drawing and Technical Specification Section 700 and 2500	cum	As Per Annexure - 8.04,8.05 & 8.10				1,44,113
8.06	Seeding and Mulching of Cut Slopes / Embankment Slopes complete as per drawings and Technical Specification Clause 308	sqm	As Per Annexure - 2.1				4,40,114
8.07	Geo-synthetic Mat for Erosion Control complete as per drawings and Technical Specification clause	sqm	As Per Annexure - 2.1				78,449
8.08	Soil Nailing complete as per drawings and Technical Specification Section 3200	m	As Per Annexure - 2.1	60415	6 / 4 (average)		90,623
8.09	Temporary Facing complete as per drawings and Technical Specification 3200.	sqm	As Per Annexure - 2.1				7,815
8.10	Reinforced Soil Wall complete as per drawing and Technical Specification Section 3100.						
a)	Facia panels as per Technical Specification Section 3105 including soil reinforcing element, foundation pad, coping beam, all accessories, consumables and components of drainage system (filter media, drainage layer, drain pipe, catch pit etc.), including ground improvement complete.	sqm	As Per Annexure - 8.04,8.05 & 8.10				187
b)	Excavation for RE wall as per drawings and Technical Specification, in						
i	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	As Per Annexure - 8.04,8.05 & 8.10				4,261
ii	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum					
iii	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum					
iv	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	As Per Annexure - 8.04,8.05 & 8.10				1,826
c)	Back filling behind RE Wall Complete as per drawings and Technical Specification Clause 305.	cum	As Per Annexure - 8.04,8.05 & 8.10				12,533
8.11	Construction of Parapet Wall in RR Masonry (CM 1:3) as per Drawings and Technical Specification	cum	1	23795	0.450	0.700	7,495

Project Road : Pfutsero Phek Road

Annexure 2.1

BOQ Item No.

2.01 Project Road Locations not covered under BoQ Item 2.02, 2.03 and 2.04

Cut Quantity	36,59,318	(Summation of Volume under Col. 2 of Annexure 2.1 (d) minus Total Subgrade shape volume under Col. 10)
--------------	-----------	---

2.02 Project Road Locations having Sand Stone away by more than 200 m of habitated area

From	To	Cut Quantity	Remarks
8070	8130	6,867	Summation of respective Volumes under Col. 2 of Annexure 2.1 (d)
19070	19435	11,023	
19685	19885	9,668	
20240	20440	31,443	
21660	21860	1,894	
26475	27380	1,66,509	
31240	31830	1,47,170	
34360	34755	84,492	
53650	53920	15,878	
Total		4,77,942	

2.03 Project Road Locations having Sand Stone within 200 m of habitated area

From	To	Cut Quantity	Remarks
7990	8070	36,521	Summation of respective Volume under Col. 2 of Annexure 2.1 (d)

2.04 Project Road Locations having Sand Stone within 20 m of habitated area

From	To	Cut Quantity	Remarks
		NIL	

2.05 Embankment Construction

1,49,325 (Summation of Volumes under Col. 3 of Annexure 2.1 (d))

2.06 Subgrade Preparation from Borrow Earth (CBR > Designed CBR)

1,00,075 (25% of Summation of Volumes under Col. 10 of Annexure 2.1(d))

2.07 Preparation of Subgrade in Cut Formation (Loosening and Compacting to full depth)

2,90,430 (75% of Summation of Volumes under Col. 10 of Annexure 2.1(d))

2.08 Preparation of Subgrade in Rocky Formation

19,590 Surface Area for locations having Sand stone Rock

3.01 GSB

1,59,305 summation of Volumes in Col. 9 in Annexure 2.1 (d)

3.02 WMM

1,82,037 summation of Volumes in Col. 8 in Annexure 2.1 (d)

4.01 Prime Coat

7,08,270 summation of Surface Area in Col. 7 in Annexure 2.1 (d)

4.02 Tack Coat

7,04,907 summation of Surface Area in Col. 5 in Annexure 2.1 (d)

4.03 DBM

35,413 summation of Volumes in Col. 6 in Annexure 2.1 (d)

4.04 BC

21,147 summation of Volumes in Col. 4 in Annexure 2.1 (d)

8.06 Seeding and Mulching

4,40,114 summation of Cutslope Surface Areas for Category II & III stretches as per slope stability surveys

8.07 Geo-synthetic Mat

75,514 summation of Cutslope Surface Areas under Landslide prone stretches

8.08 Nailing

60,415 summation of Cutslope Surface Areas under Landslide prone stretches

8.09 Temporary Facing

7,815 summation of Surface Areas for first Cut height in Landslide prone stretches

Annexure 2.1 (d) : Template Shape / Component Report (from Mx Roads) : PP Road											Cut Slope			
Station	Cut Volume	Fill Volume	BC Volume	Tack Coat Sur. Area	DBM Volume	Prime Coat Sur. Area	WMM Volume	GSB Volume	Subgrade Volume	<6m Sur. Area	<12 m Sur. Area	> 12 m Sur. Area	R Wall Height	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	
0														
10	129.1	5.9	3.1	103.3	5.2	104.0	26.8	23.4	54.1	11.0			1.7	
20	115.1	12.5	3.0	100.0	5.0	100.0	25.5	21.9	54.1	15.0			1.9	
30	126.8	13.2	3.0	100.0	5.0	100.0	25.5	21.9	54.1	29.0			1.7	
36	81.7	6.6	1.7	56.7	2.9	58.0	14.5	12.5	30.7	24.0			1.6	
40	65.7	4.3	1.3	43.3	2.2	44.0	11.0	9.5	23.3	21.0			1.5	
50	175.0	8.2	3.1	103.3	5.2	104.0	26.2	22.5	55.3	58.0			1.3	
51	13.5	0.5	0.2	6.7	0.4	8.0	1.8	1.6	3.9	4.0			1.3	
55	99.1	3.3	1.4	46.7	2.4	48.0	12.3	10.5	25.8		35.0		1.3	
60	116.8	3.6	1.6	53.3	2.6	52.0	13.3	11.4	28.0		76.0		1.4	
70	202.7	10.9	3.3	110.0	5.5	110.0	27.7	23.8	58.3	31.0			1.8	
80	144.5	15.4	3.3	110.0	5.5	110.0	27.7	23.8	58.3	54.0			2.0	
88	66.2	13.9	2.6	86.7	4.3	86.0	21.6	18.5	45.5	25.0			2.0	
90	8.5	4.6	0.7	23.3	1.2	24.0	5.9	5.1	12.5	3.0			2.1	
92	7.8	5.9	0.7	23.3	1.2	24.0	6.2	5.3	13.1	2.0			2.2	
100	22.4	27.1	2.4	80.0	3.9	78.0	20.0	17.1	42.2	11.0			2.7	
107	21.6	26.2	2.2	73.3	3.7	74.0	18.7	16.1	39.6	11.0			2.2	
110	8.2	8.3	0.8	26.7	1.3	26.0	6.8	5.9	14.5	2.0			2.4	
120	25.2	48.1	3.0	100.0	5.0	100.0	25.5	21.9	54.0	4.0			3.1	
130	25.5	54.6	3.0	100.0	5.0	100.0	25.5	21.9	54.0	5.0			2.8	
140	59.9	30.7	3.0	100.0	5.0	100.0	25.5	21.9	53.9	22.0			1.7	
150	106.3	10.3	3.0	100.0	5.0	100.0	25.5	21.9	53.9	45.0			1.2	
160	185.7	6.2	3.1	103.3	5.1	102.0	26.0	22.3	54.7		74.0		1.0	
161	22.5	0.5	0.3	10.0	0.5	10.0	2.4	2.0	5.0		13.0		1.0	
161	2.6	0.1			0.1	2.0	0.3	0.2	0.6		2.0		1.3	
161													1.3	
170	222.6	10.1	2.9	96.7	4.8	96.0	24.5	21.0	51.7		137.0		1.7	
176	101.5	9.3	2.0	66.7	3.3	66.0	16.6	14.3	35.0	14.0			2.0	
180	39.5	8.2	1.3	43.3	2.2	44.0	11.1	9.5	23.4	17.0			2.2	
180	4.0	1.2	0.2	6.7	0.3	6.0	1.4	1.2	2.9	2.0			2.2	
190	91.9	17.0	3.1	103.3	5.2	104.0	26.4	22.6	55.4	38.0			1.7	
200	114.3	13.7	3.4	113.3	5.7	114.0	29.0	25.4	64.1	38.0				
210	101.6	8.1	3.5	116.7	5.8	116.0	30.4	27.0	69.9	31.0				
220	80.9	0.4	3.5	116.7	5.8	116.0	30.4	27.0	69.9	19.0				
230	84.2	0.4	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
240	96.3	0.7	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
250	99.0	0.5	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
260	96.1	0.2	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
270	91.2	0.8	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
280	87.6	1.2	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
290	69.5	4.6	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
300	50.1	14.2	3.5	116.7	5.8	116.0	30.4	27.0	69.9					
302	8.9	3.8	0.6	20.0	1.1	22.0	5.6	5.0	12.8					
310	40.0	14.9	2.8	93.3	4.8	96.0	24.8	22.0	56.9					
320	51.9	14.0	3.5	116.7	5.8	116.0	30.3	26.9	69.3					
328	44.2	13.0	2.8	93.3	4.6	92.0	24.1	21.3	54.9					
330	11.2	4.3	0.7	23.3	1.2	24.0	6.2	5.5	14.1					
340	52.9	22.3	3.5	116.7	5.8	116.0	30.2	26.8	68.8					
350	57.7	16.6	3.5	116.7	5.8	116.0	30.2	26.8	68.8					
360	78.6	7.4	3.5	116.7	5.8	116.0	30.2	26.8	68.8	1.0				
370	107.8	5.0	3.4	113.3	5.7	114.0	29.0	25.3	58.6	4.0			1.0	
380	100.0	11.1	3.3	110.0	5.5	110.0	27.8	23.8	58.6	5.0			2.1	
390	77.7	30.9	3.3	110.0	5.5	110.0	27.8	23.8	58.6	6.0			3.5	
400	79.9	48.0	3.3	110.0	5.5	110.0	27.8	23.8	58.6	8.0			3.7	
410	80.6	53.5	3.3	110.0	5.5	110.0	28.0	24.0	59.0	8.0			3.8	
413	21.2	16.7	0.9	30.0	1.5	30.0	7.8	6.7	16.5	2.0			4.0	
413	0.7	0.6			0.1	2.0	0.3	0.2	0.6				3.9	
413													3.9	
420	43.8	46.3	2.4	80.0	4.0	80.0	20.2	17.3	42.5	3.0			4.2	
428	38.0	59.7	2.6	86.7	4.4	88.0	22.4	19.2	47.2	1.0			4.8	
430	8.8	18.2	0.7	23.3	1.2	24.0	6.1	5.2	12.7				5.0	
440	33.5	77.0	3.4	113.3	5.6	112.0	28.5	24.4	59.8				4.4	
440	1.0	2.4	0.1	3.3	0.2	4.0	1.1	0.9	2.2				4.3	
450	26.4	37.0	3.2	106.7	5.4	108.0	27.4	23.5	57.5				1.9	
460	40.3	9.8	3.4	113.3	5.6	112.0	28.5	24.4	59.7	2.0			1.0	
470	75.8	5.6	3.4	113.3	5.6	112.0	28.6	24.5	59.9	12.0			1.0	
479	95.7	4.9	3.0	100.0	5.0	100.0	25.1	21.5	52.6	17.0			1.0	
479	1.3	0.1			0.1	2.0	0.3	0.2	0.6				1.2	
479													1.5	
480	16.7	1.6	0.4	13.3	0.7	14.0	3.4	2.9	7.2	3.0			1.5	
490	130.8	15.0	3.4	113.3	5.7	114.0	29.1	24.9	61.2	26.0			1.6	
500	103.5	20.5	3.5	116.7	5.8	116.0	29.3	25.1	61.6	21.0			1.7	
510	72.8	28.6	3.5	116.7	5.8	116.0	29.3	25.1	61.6	16.0			1.9	
520	51.2	37.0	3.5	116.7	5.8	116.0	29.3	25.1	61.6	10.0			2.2	
530	36.5	45.6	3.5	116.7	5.8	116.0	29.3	25.1	61.6	6.0			2.5	
540	26.4	54.6	3.5	116.7	5.8	116.0	29.3	25.1	61.6	4.0			2.9	
550	18.8	64.4	3.5	116.7	5.8	116.0	29.3	25.1	61.6	2.0			3.2	
560	14.1	72.1	3.5	116.7	5.8	116.0	29.3	25.1	61.6	1.0			3.2	
570	11.4	76.8	3.5	116.7	5.8	116.0	29.3	25.1	61.6				3.3	
580	16.8	79.6	3.5	116.7	5.8	116.0	29.3	25.1	61.6	4.0			3.5	
590	38.6	75.4	3.5	116.7	5.8	116.0	29.3	25.1	61.6	14.0			3.5	
592	12.9	15.3	0.8	26.7	1.3	26.0	6.7	5.7	14.0	5.0			3.3	
600	60.7	37.0	2.7	90.0	4.5	90.0	22.6	19.4	47.5	19.0			2.4	
607	91.6	17.8	2.5	83.3	4.2	84.0	21.3	18.4	44.8	23.0			1.6	
610	44.5	4.3	0.9	30.0	1.6	32.0	8.0	6.8	16.8	10.0			1.6	
620	196.7	13.1	3.5	116.7	5.8	116.0	29.3	25.1	61.7	39.0			1.4	
621	17.5	0.9	0.3	10.0	0.5	10.0	2.3	2.0	4.8	3.0			1.4	
630	257.6	9.5	3.3	110.0	5.5	110.0	28.1	24.5	61.5	44.0				
640	389.2	4.8	3.6	120.0	6.0	120.0	30.7	27.0	63.6	58.0				
650	491.8	2.7	3.5	116.7	5.8	116.0	29.5	25.6	63.6		82.0		1.0	
653	186.7	2.0	1.2	40.0	2.0	40.0	10.1	8.7	21.3		58.0		1.0	
660	355.2	8.4	2.3	76.7	3.8	76.0	19.2	16.4	40.4		117.0		1.9	
670	509.3	19.9	3.5	116.7	5.8	116.0	29.3	25.1	61.6		144.0		1.7	
680	538.0	15.1	3.5	116.7	5.8	116.0	29.3	25.1	61.5		203.0		1.3	
685	331.4	4.5	1.9	63.3	3.1	62.0	15.7	13.4	32.9		166.0		1.2	
690	312.1	3.1	1.6	53.3	2.7	54.0	13.6	11.6	28.4		148.0		1.1	
700	706.3	6.2	3.5	116.7	5.8	116.0	29.2	25.0	61.3		331.0		1.1	
710	661.2	7.5	3.5	116.7	5.8	116.0	29.2	25.0	61.3		354.0		1.7	
720	613.9	16.0	3.5	116.7	5.8	116.0	29.2	25.0	61.3		290.0		2.4	
726	416.8	11.6	2.0	66.7	3.3	66.0	16.7	14.3	35.0			150.0	2.1	
730	375.7	6.1	1.5	50.0	2.5	50.0	12.6	10.8	26.3			229.0	1.7	
740	1083.2	7.8	3.5	116.7	5.8	116.0	29.2	25.0	61.3			418.0	0.9	
750	1587.7	3.5	3.5	116.7	5.8	116.0	29.2	25.0	61.2			506.0	0.5	
756	1061.8	2.1	2.0	66.7	3.3	66.0	16.7	14.3	34.9			412.0	1.1	
760	706.5	3.2	1.5	50.0	2.5	50.0								

Annexure 2.1 (d) : Template Shape / Component Report (from Mx Roads) : PP Road											Cut Slope			
Station	Cut Volume	Fill Volume	BC Volume	Tack Coat Sur. Area	DBM Volume	Prime Coat Sur. Area	WMM Volume	GSB Volume	Subgrade Volume	<6m Sur. Area	<12 m Sur. Area	> 12 m Sur. Area	R Wall Height	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	
58060	440.3	12.4	1.7	56.7	2.8	56.0	14.8	13.2	34.1		45.0			
58070	595.8	35.0	3.1	103.3	5.2	104.0	26.6	23.3	53.6		141.0		2.5	
58070	6.3	0.5			0.1	2.0	0.3	0.3	0.7		1.0		2.5	
58080	401.3	68.4	3.1	103.3	5.2	104.0	26.8	23.4	59.4		140.0			
58090	276.3	96.6	3.2	106.7	5.4	108.0	28.0	24.9	64.6	32.0				
58092	31.3	13.4	0.5	16.7	0.8	16.0	4.2	3.8	9.9	9.0				
58100	165.0	70.9	2.7	90.0	4.6	92.0	23.8	21.2	55.2	48.0				
58110	180.8	55.1	3.2	106.7	5.4	108.0	28.0	24.9	64.6	43.0				
58120	165.4	12.1	3.2	106.7	5.4	108.0	28.0	24.9	64.6	31.0				
58130	135.1		3.3	110.0	5.5	110.0	28.7	25.5	66.0	21.0				
58131	7.6		0.2	6.7	0.4	8.0	1.9	1.7	4.5	1.0				
58140	135.9	0.1	3.1	103.3	5.2	104.0	27.0	23.8	55.8	11.0				
58150	170.3	0.1	3.4	113.3	5.7	114.0	29.3	25.8	65.8	11.0				
58160	146.6		3.5	116.7	5.8	116.0	30.3	26.8	69.1	7.0				
58170	146.3	1.0	3.4	113.3	5.7	114.0	29.3	25.8	60.5	6.0				
58180	160.4	1.1	3.4	113.3	5.7	114.0	29.3	25.8	65.8	6.0				
58187	109.4	0.1	2.5	83.3	4.2	84.0	21.6	19.1	49.3	3.0				
58190	40.0		1.0	33.3	1.7	34.0	8.7	7.7	19.8	1.0				
58200	103.9	0.1	3.5	116.7	5.8	116.0	30.3	26.9	69.3	3.0				
58202	17.0		0.7	23.3	1.3	26.0	6.5	5.8	14.9					
58210	77.2		2.8	93.3	4.7	94.0	24.2	21.5	55.6	3.0				
58213	35.2		1.1	36.7	1.8	36.0	9.2	8.2	19.0	3.0				
58220	98.9		2.3	76.7	3.9	78.0	20.0	17.5	44.1	16.0				
58230	190.3		3.5	116.7	5.8	116.0	30.1	26.5	67.8	44.0				
58240	184.9		3.6	120.0	6.0	120.0	31.2	27.7	71.7	53.0				
58250	181.3		3.6	120.0	6.0	120.0	31.2	27.7	71.7	57.0				
58257	145.3		2.5	83.3	4.2	84.0	22.0	19.6	50.7	45.0				
58260	60.1	0.1	1.0	33.3	1.8	36.0	9.2	8.2	21.1	18.0				
58270	182.0	3.8	3.4	113.3	5.8	116.0	29.7	25.9	59.6	55.0			1.1	
58280	267.2	5.3	3.4	113.3	5.7	114.0	29.5	25.7	64.6		94.0			
58290	550.8	1.8	3.5	116.7	5.9	118.0	30.4	27.0	69.4	262.0				
58293	209.7		0.9	30.0	1.6	32.0	8.0	7.1	18.3	92.0				
58300	671.3	0.1	2.5	83.3	4.3	86.0	22.2	19.6	50.5	270.0				
58310	1103.8	0.1	3.5	116.7	5.8	116.0	30.2	26.8	68.8		259.0			
58318	843.4	1.5	2.7	90.0	4.6	92.0	23.7	21.0	53.9		303.0			
58320	196.5	1.1	0.8	26.7	1.3	26.0	6.6	5.8	14.9		85.0			
58330	603.7	9.7	3.4	113.3	5.7	114.0	29.0	25.3	58.6	45.0		1.5		
58340	267.9	22.6	3.3	110.0	5.5	110.0	27.8	23.8	58.5	86.0		1.9		
58343	120.4	10.0	1.0	33.3	1.6	32.0	8.3	7.1	17.4		79.0	1.9		
58350	479.4	28.2	2.3	76.7	3.8	76.0	19.5	16.7	41.0		376.0	2.0		
58360	827.4	52.8	3.3	110.0	5.5	110.0	27.7	23.8	58.3		550.0	2.1		
58370	917.3	63.2	3.3	110.0	5.5	110.0	27.7	23.8	58.3		558.0	2.2		
58380	902.5	66.3	3.3	110.0	5.5	110.0	27.7	23.8	58.3		551.0	2.3		
58387	560.5	46.3	2.2	73.3	3.7	74.0	18.5	15.9	38.9		353.0	2.4		
58390	268.6	25.8	1.1	36.7	1.9	38.0	9.6	8.2	20.1		173.0	2.5		
58400	677.4	69.1	3.4	113.3	5.6	112.0	28.6	24.5	60.1		85.0	2.2		
58410	693.8	43.4	3.4	113.3	5.7	114.0	28.9	24.7	60.7		256.0	2.2		
58416	466.8	15.0	1.9	63.3	3.3	66.0	16.5	14.1	34.7		193.0	2.1		
58420	380.3	16.3	1.6	53.3	2.6	52.0	13.3	11.6	29.2		149.0			
58430	869.1	31.5	3.6	120.0	6.1	122.0	31.7	28.0	71.6		336.0			
58440	794.0	32.0	3.6	120.0	6.1	122.0	31.7	28.0	71.6		328.0			
58446	378.7	28.5	2.1	70.0	3.6	72.0	18.6	16.5	42.1		125.0			
58450	207.6	14.0	1.5	50.0	2.5	50.0	12.6	10.9	25.4		61.0	1.7		
58460	415.1	13.9	3.4	113.3	5.7	114.0	29.0	24.9	61.0		179.0	1.1		
58470	392.9	3.2	3.5	116.7	5.9	118.0	30.1	26.3	66.2		165.0			
58480	407.5	0.4	3.6	120.0	6.0	120.0	31.3	27.8	71.8		159.0			
58487	271.8	0.3	2.4	80.0	4.1	82.0	21.4	19.1	49.3	21.0				
58490	115.5	0.1	1.1	36.7	1.9	38.0	9.7	8.7	22.4	19.0				
58500	315.9	0.3	3.6	120.0	6.0	120.0	31.2	27.7	71.8	57.0				
58510	247.3	0.4	3.6	120.0	6.0	120.0	31.2	27.7	71.8	42.0				
58520	186.5	1.5	3.6	120.0	6.0	120.0	31.2	27.7	71.8	28.0				
58530	152.1	6.9	3.6	120.0	6.0	120.0	31.2	27.7	71.8	18.0				
58540	150.8	6.5	3.6	120.0	6.0	120.0	31.2	27.7	71.8	18.0				
58550	168.5	4.6	3.6	120.0	6.0	120.0	31.2	27.7	71.8	38.0				
58554	88.1	5.3	1.6	53.3	2.7	54.0	13.8	12.3	31.8	24.0				
58560	113.4	6.5	1.9	63.3	3.2	64.0	16.4	14.3	32.8	32.0			1.1	
58567	131.1	5.9	2.2	73.3	3.7	74.0	18.8	16.1	39.6	37.0			1.4	
58570	55.9	3.5	1.0	33.3	1.6	32.0	8.1	7.0	17.1	17.0			1.5	
58580	183.2	13.6	3.0	100.0	5.1	102.0	25.9	22.2	54.6	52.0			1.5	
58582	35.0	2.8	0.6	20.0	1.0	20.0	4.9	4.2	10.4	9.0			1.8	
58590	140.8	12.7	2.4	80.0	4.1	82.0	20.6	17.7	43.6	38.0			2.0	
58600	173.6	15.8	3.0	100.0	5.0	100.0	25.5	21.9	54.1	55.0			1.6	
58610	186.4	9.9	3.0	100.0	5.0	100.0	25.5	21.9	54.1	57.0			1.1	
58620	192.7	5.4	3.0	100.0	5.0	100.0	25.5	21.9	54.1	56.0			0.9	
58630	172.9	6.0	3.0	100.0	5.0	100.0	25.5	21.9	54.1	51.0			1.2	
58635	74.3	4.2	1.5	50.0	2.5	50.0	12.6	10.8	26.6	23.0			1.3	
58640	68.1	6.5	1.5	50.0	2.6	52.0	13.0	11.1	27.4	23.0			1.5	
58650	103.3	21.5	3.1	103.3	5.3	106.0	27.2	23.8	60.2	43.0				
58650	0.6	0.2					0.2	0.2	0.5					
58657	65.3	15.1	2.5	83.3	4.1	82.0	21.1	18.5	42.5	33.0			1.2	
58660	27.5	4.2	0.8	26.7	1.4	28.0	7.1	6.1	15.0	13.0			1.4	
58670	185.9	18.9	3.3	110.0	5.5	110.0	27.7	23.8	58.3		84.0		1.8	
58676	162.2	11.9	1.9	63.3	3.3	66.0	16.5	14.1	34.7		99.0		2.0	
58680	129.9	6.9	1.3	43.3	2.2	44.0	11.1	9.5	23.3		66.0		1.8	
58683	116.4	4.2	1.1	36.7	1.8	36.0	9.3	8.0	19.5		54.0		1.8	
58690	200.8	9.1	2.0	66.7	3.4	68.0	17.1	14.7	36.1	21.0			1.7	
58698	243.8	9.9	2.5	83.3	4.3	86.0	21.6	18.6	45.8	53.0			1.2	
58700	44.2	1.2	0.5	16.7	0.8	16.0	3.9	3.4	8.4	9.0			1.2	
58710	287.5	6.6	3.0	100.0	5.0	100.0	25.5	21.9	54.1	55.0			0.9	
58720	280.1	5.6	3.0	100.0	5.0	100.0	25.5	21.9	54.1	49.0			1.0	
58727	184.5	3.2	2.1	70.0	3.6	72.0	18.4	16.1	40.7	32.0				
58730	87.2	1.0	1.0	33.3	1.7	34.0	8.8	7.8	20.3	15.0				
58740	279.9	3.0	3.2	106.7	5.4	108.0	28.0	24.9	64.6	48.0				
58750	262.2	8.6	3.2	106.7	5.4	108.0	28.0	24.9	64.6	49.0				
58757	174.8	7.3	2.3	76.7	3.8	76.0	19.5	17.0	39.3	36.0			1.1	
58760	62.0	1.8	0.8	26.7	1.4	28.0	7.0	6.0	14.7	14.0			1.1	
58770	214.7	7.3	3.1	103.3	5.1	102.0	26.1	22.4	55.0	49.0			1.2	
58772	45.4	1.9	0.7	23.3	1.2	24.0	6.1	5.2	12.8	11.0			1.2	
58777	85.7	4.1	1.4	46.7	2.4	48.0	12.3	10.5	25.8	21.0			1.3	
58780	60.1	3.8	1.1	36.7	1.8	36.0	8.9	7.7	18.8	15.0			1.4	
58790	168.2	20.7	3.3	110.0	5.5	110.0	27.7	23.8	58.3	42.0			1.8	
58800	118.6	35.7	3.3	110.0										

QUANTITY CALCULATION FOR 1X1.2 DIA PIPES

S. No.	Proposed Chainage (km)	Designation	Pipe Length	Excavation	Back Filling	Conc. Foundation	Conc. Substructure	Levelling coarse	Filter Media	Painting	Lettering
1	0.925	1*1.2	12.500	102.174	137.214	12.96	20.4	3.472	30.24	2.28	50
2	8.050	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
3	30.940	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
4	46.415	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
5	47.050	1*1.2	12.500	165.490	179.180	14.58	24	3.875	33.84	2.28	50
6	57.430	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
7	43.950	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
8	49.670	1*1.2	12.500	137.415	137.214	12.96	20.4	3.472	30.24	2.28	50
9	15.150	1*1.2	12.500	137.415	137.214	12.96	20.4	3.472	30.24	2.28	50
10	33.510	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
11	48.965	1*1.2	12.500	165.490	179.180	14.58	24	3.875	33.84	2.28	50
12	60.595	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
13	61.820	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
14	54.240	1*1.2	12.500	137.779	137.214	12.96	20.4	3.472	30.24	2.28	50
15	57.770	1*1.2	10.000	137.144	137.214	12.96	20.4	3.472	30.24	2.28	50
16	57.840	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
17	30.760	1*1.2	12.500	133.214	137.214	12.96	20.4	3.472	30.24	2.28	50
18	12.910	1*1.2	12.500	163.543	179.180	14.58	24	3.875	33.84	2.28	50
19	22.800	1*1.2	12.500	143.779	137.214	12.96	20.4	3.472	30.24	2.28	50
20	50.510	1*1.2	12.500	132.630	137.214	12.96	20.4	3.472	30.24	2.28	50
21	12.210	1*1.2	12.500	129.215	137.214	12.96	20.4	3.472	30.24	2.28	50
22	50.475	1*1.2	12.500	127.812	137.214	12.96	20.4	3.472	30.24	2.28	50
23	45.640	1*1.2	12.500	127.604	137.214	12.96	20.4	3.472	30.24	2.28	50
24	52.970	1*1.2	10.000	131.343	137.214	12.96	20.4	3.472	30.24	2.28	50
25	59.630	1*1.2	12.500	125.924	137.214	12.96	20.4	3.472	30.24	2.28	50

S. No.	Proposed Chainage (km)	Designation	Pipe Length	Excavation	Back Filling	Conc. Foundation	Conc. Substructure	Levelling coarse	Filter Media	Painting	Lettering
--------	------------------------	-------------	-------------	------------	--------------	------------------	--------------------	------------------	--------------	----------	-----------

QUANTITY CALCULATION FOR 2X1.2 DIA PIPES

236	27.490	2*1.2	25.000	132.630	137.214	18.150	21.74279674	3.596	30.24	2.28	50
237	5.500	2*1.2	25.000	132.567	137.214	18.150	21.74279674	3.596	30.24	2.28	50
238	41.300	2*1.2	25.000	127.608	137.214	18.150	21.74279674	3.596	30.24	2.28	50
239	21.450	2*1.2	25.000	127.185	137.214	18.150	21.74279674	3.596	30.24	2.28	50
240	2.950	2*1.2	25.000	157.743	179.180	20.525	26.24279674	3.968	33.84	2.28	50
241	3.260	2*1.2	20.000	128.347	137.214	18.150	21.74279674	3.596	30.24	2.28	50
242	3.200	2*1.2	25.000	127.461	137.214	18.150	21.74279674	3.596	30.24	2.28	50
243	3.122	2*1.2	25.000	125.367	137.214	18.150	21.74279674	3.596	30.24	2.28	50
244	13.720	2*1.2	25.000	120.233	137.214	18.150	21.74279674	3.596	30.24	2.28	50
245	3.680	2*1.2	25.000	148.032	179.180	20.525	26.24279674	3.968	33.84	2.28	50
246	6.165	2*1.2	25.000	118.663	137.214	18.150	21.74279674	3.596	30.24	2.28	50
247	16.760	2*1.2	25.000	117.945	137.214	18.150	21.74279674	3.596	30.24	2.28	50
248	4.445	2*1.2	25.000	145.396	179.180	20.525	26.24279674	3.968	33.84	2.28	50
249	21.555	2*1.2	25.000	116.588	137.214	18.150	21.74279674	3.596	30.24	2.28	50
250	25.920	2*1.2	25.000	116.475	137.214	18.150	21.74279674	3.596	30.24	2.28	50
251	44.465	2*1.2	25.000	113.938	137.214	18.150	21.74279674	3.596	30.24	2.28	50
252	4.090	2*1.2	25.000	140.290	179.180	20.525	26.24279674	3.968	33.84	2.28	50
253	17.000	2*1.2	25.000	111.405	137.214	18.150	21.74279674	3.596	30.24	2.28	50
254	6.820	2*1.2	25.000	137.753	179.180	20.525	26.24279674	3.968	33.84	2.28	50

S. No.	Proposed Chainage	Designation	Excavation	RCC for box	Approach slab RCC	Exp. Gap	Lettering	Drainage Spout	Painting	Levelling Course	Excavation for Return Wall	RCC for Return Wall	Filter Media	Backfill	RCC for Counter fort	RCC for Wall
	(km)		m ³	m ³	m ³	lm	nos.	nos.	m ²	m ³	m ³	m ³	m ³	m ³	m ³	m ³
17	45.685	1.22.00	90.09	56.51	26.71	21.80	50.00	4.00	9.88	15.87	70.20	22.24	20.16	91.48	0.00	0.00
18	40.465	1.22.00	89.34	56.51	26.71	21.80	50.00	4.00	9.88	15.87	69.83	22.24	20.16	91.48	0.00	0.00
19	55.970	1.22.00	91.70	57.95	27.44	22.40	50.00	4.00	9.88	16.30	69.78	22.24	20.16	91.48	0.00	0.00
20	1.260	1.22.00	91.43	57.95	27.44	22.40	50.00	4.00	9.88	16.30	69.65	22.24	20.16	91.48	0.00	0.00
21	2.090	1.22.00	88.01	56.51	26.71	21.80	50.00	4.00	9.88	15.87	69.16	22.24	20.16	91.48	0.00	0.00
22	39.260	1.22.00	84.70	55.07	25.97	21.20	50.00	4.00	9.88	15.44	68.70	22.24	20.16	91.48	0.00	0.00
23	22.470	1.22.00	87.89	59.39	28.18	23.00	50.00	4.00	9.88	16.73	66.79	22.24	20.16	91.48	0.00	0.00
24	47.870	1.22.00	86.67	56.51	26.71	21.80	50.00	4.00	9.88	15.87	68.48	22.24	20.16	91.48	0.00	0.00
25	51.140	1.22.00	91.18	59.39	28.18	23.00	50.00	4.00	9.88	16.73	68.36	22.24	20.16	91.48	0.00	0.00
26	42.250	1.22.00	86.20	56.51	26.71	21.80	50.00	4.00	9.88	15.87	68.25	22.24	20.16	91.48	0.00	0.00
27	47.300	1.22.00	90.92	59.39	28.18	23.00	50.00	4.00	9.88	16.73	68.23	22.24	20.16	91.48	0.00	0.00
28	52.560	1.22.00	88.19	57.86	27.39	22.36	50.00	4.00	9.88	16.27	68.14	22.24	20.16	91.48	0.00	0.00
29	54.110	1.22.00	78.30	52.19	24.50	20.00	50.00	4.00	9.88	14.59	67.82	22.24	20.16	91.48	0.00	0.00
30	48.180	1.22.00	84.87	56.51	26.71	21.80	50.00	4.00	9.88	15.87	67.58	22.24	20.16	91.48	0.00	0.00
31	44.165	1.22.00	90.80	61.02	29.01	23.68	50.00	4.00	9.88	17.22	66.94	22.24	20.16	91.48	0.00	0.00
32	2.560	1.22.00	80.07	55.07	25.97	21.20	50.00	4.00	9.88	15.44	66.31	22.24	20.16	91.48	0.00	0.00

SL. No.	Proposed Chainage	DOWNSTREAM PROTECTION WORKS										UPSTREAM PROTECTION WORKS									
		Excavation	PCC M15	RCC for flooring+ww all +block	Rcc in Front and Back Curtain Wall	RCC Total	Boulders in Wire Crates	Reinforce ment	Excavation	Excavation for Catchpit	Excavation Total	PCC M15	RCC	RCC for Catchpit	RCC Total	Boulders in Wire Crates	Reinforce ment	Reinforce ment for Catchpit	Reinforce ment Total		
	(km)	m³	m²	m³	m³	m³	m³	MT	m³	m³	m³	m²	m³	m³	m³	m³	MT	MT	MT		
82	44.625	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
83	56.960	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
84	55.655	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
85	19.890	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
86	25.500	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
87	23.970	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
88	52.510	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
89	12.790	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
90	7.040	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
91	0.630	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
92	46.485	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
93	21.755	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
94	49.960	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
95	46.960	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
96	59.470	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
97	29.810	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
98	60.845	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
99	54.845	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
100	60.430	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
101	20.370	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
102	24.120	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
103	30.515	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
104	12.410	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
105	55.240	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
106	8.470	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
107	32.180	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
108	31.920	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
109	28.135	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
110	51.540	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
111	41.120	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
112	19.710	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
113	46.130	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
114	48.140	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
115	57.250	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
116	40.775	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
117	9.710	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
118	58.780	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
119	49.020	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
120	20.100	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
121	0.480	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		
122	40.705	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20		

QUANTITIES FOR PROTECTION WORK FOR 2.0m x2.0m BOX CULVERTS

SL. No.	Proposed Channage	DOWNSTREAM PROTECTION WORKS										UPSTREAM PROTECTION WORKS									
		Excavation	PCC M15	RCC for flooring+ wall +block	Rcc in Front and Back Curtain Wall	RCC Total	Boulders in Wire Crates	Reinforce ment	Excavation	Excavation for Catchpit	Excavatio n Total	PCC M15	RCC	RCC For Catchpit	RCC Total	Boulders in Wire Crates	Reinforce ment	Reinforce ment for Catchpit	Reinforce ment Total		
	(km)	m ³	m ³	m ³	m ³	m ³	MT	m ³	m ³	m ³	m ³	m ³	m ³	m ³	m ³	MT	MT	MT	MT		
1	31.565	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
2	47.590	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
3	44.510	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
4	25.160	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
5	51.600	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
6	8.840	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
7	57.490	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
8	44.030	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
9	8.710	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
10	44.140	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
11	49.520	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
12	44.335	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
13	25.430	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
14	41.485	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
15	51.420	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
16	53.755	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
17	45.685	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
18	40.465	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
19	55.970	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
20	1.260	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
21	2.090	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
22	39.280	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
23	22.470	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
24	47.870	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
25	51.140	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
26	42.260	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
27	47.300	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
28	52.560	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
29	54.110	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
30	48.180	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
31	44.165	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
32	2.580	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
33	48.670	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
34	46.725	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		
35	42.715	50.93	3.12	6.98	5.39	12.37	32.40	0.62	61.64	32.27	93.91	6.36	22.12	3.01	25.13	16.20	1.15	0.24	1.39		

**Detailed Calculation For Upstream Protection Works for 2mx2m Single cell Box
Culvert**

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m ²)	H (m) / Thick. (m)	Volume/Qty (m ³)	SUB HEAD
	Segment 1									
5.01	Earthwork in excavations:									
	For RCC Rigid flooring (M-30)		m ³	1.00	3.00	4.40	13.20	0.60	7.92	7.92
	For Hand woven wire crates in flexible apron for protection work in upstream		m ³	1.00	3.00	5.40	16.20	1.30	21.06	21.06
	For curtain wall in segment 1		m ³	1.00	5.40	1.70		1.40	12.85	12.85
									Sub Total	41.83
	70% in Ordinary Soil									29.28
	30% in Hard Rock									12.55
5.02	PCC M-15 below curtain wall	100 mm thick.	m ³	1.00	5.40	1.70		0.10	0.92	0.92
5.02	PCC M-15 below RCC flooring	100 mm thick.	m ³	1.00	3.00	4.40	13.20	0.10	1.32	1.32
									Sub Total	2.24
5.04	RCC rigid flooring M-30			1.00	3.00	4.40	13.20	0.20	2.64	2.64
5.04	RCC Side walls over rigid flooring M-30	0.2 m thick.	m ³	2.00	3.16	0.20		1.00	1.26	1.26
									Sub Total	3.90
5.04	Rcc Curtain wall 1		m ³	1.00	5.40		0.84		4.54	4.54
5.20	Hand woven wire crates in flexible apron for	1000 mm thick.	m ³	1.00	3.00	5.40	16.20	1.00	16.20	16.20
5.07	Reinforcement in Rcc curtain wall		t	1.00	4.54	@50kg/m3 of concrete			226.80	0.227
5.07	Reinforcement in RCC flooring		t	1.00	3.90	@50kg/m3 of concrete			195.25	0.195
	Segment-2									
5.01	Earthwork in excavations:									
	Rcc slope protection structure in segment 2		m ³	1.00	1.00	3.40		0.60	2.04	2.04
	For curtain wall in segment 2		m ³	1.00	3.40	0.40		1.50	2.04	2.04
									Sub Total	2.04
	50% in Ordinary Soil									1.02
	50% in Hard rock									1.02
5.02	Pcc below Curtain wall			1.00	3.40	0.40		0.10	0.14	0.14
5.02	Pcc below Concrete for slope protection	(per meter)	m ³	1.00	1.00	3.40		0.10	0.34	0.34
									Sub Total	0.48
5.04	Rcc Concrete for slope protection structure	(per meter)	m ³	1.00	1.00	3.40		0.20	0.68	0.68
	RCC Side walls over rigid flooring M-30	(per meter)	m ³	2.00	1.00	0.20		1.00	0.40	0.40
5.04	Rcc Curtain wall		m ³	1.00	3.40	0.3		1.50	1.53	1.53
5.04	Buffle pier		m ³	28.00	0.20	0.20		0.20	0.224	0.224
									Sub Total	2.83
5.07	Reinforcement in Rcc curtain wall		t	1.00	1.5300	@50kg/m3 of concrete			76.50	0.077
5.07	Reinforcement in RCC slope protection		t	1.00	1.30	@50kg/m3 of concrete			65.20	0.065
	Segment 3									
5.01	Earthwork in excavations:									
	For catchpit		m ³	1.00	1.95	3.80	7.41	1.30	9.63	9.63
	50% in Ordinary Soil									4.82
	50% in Hard Rock									4.82
5.02	PCC M-15 below Catchpit		m ³	1.00	1.85	3.70	6.85	0.10	0.68	0.68
5.04	Rcc M-30 Catchpit walls	A1 (per meter)	m ³	2.00	1.50	0.20		1.00	0.60	0.60
		A2 (per meter)	m ³	1.00	3.40	0.20		1.00	0.68	0.68
									Sub Total	1.28
5.04	RCC M-30 Catchpit base		m ³	1.00	1.85	3.90		0.20	1.44	1.44
5.07	Reinforcement in Rcc catchpit walls		t	1.00	1.28	@50kg/m3 of concrete			64.00	0.064
	Reinforcement in Rcc catchpit base		t	1.00	1.44	@50kg/m3 of concrete			72.15	0.072

Detailed Calculation For Downstream Protection Works for 3mx3m Single cell Box Culvert

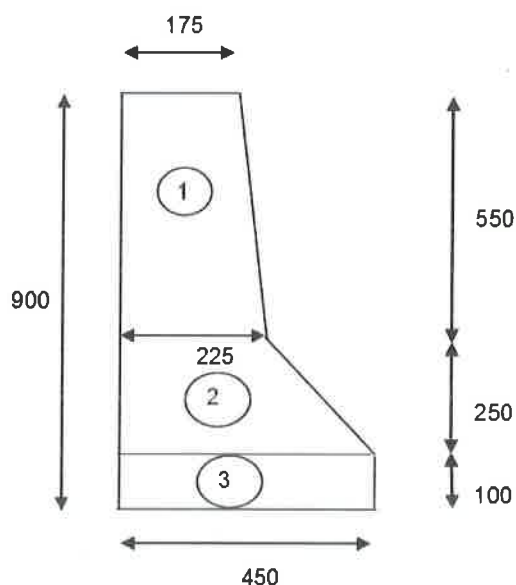
ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m ²)	H (m) / Thick. (m)	Volume/Qty (m ³)	SUB HEAD TOTAL
5.1	Earthwork in excavations for protection works									
	For RCC Rigid flooring (M-30)		m ³	1.00	5.00	5.40	27.00	0.30	8.10	8.10
	For Hand woven wire crates in flexible apron for protection work		m ³	1.00	6.00	6.40	38.40	1.00	38.40	38.40
	For front curtain wall		m ³	1.00	6.40	1.70		1.30	14.14	14.14
	For back curtain wall		m ³	0.00	4.40	0.30		1.00	0.00	0.00
	70% in soil									60.64
	30% in ordinary rock									42.45
										18.19
	Outlet/downstream Protection Work									
5.04	RCC channel/flooring M-30	0.2m thick	m ³	1.00	5.00	5.40		0.20	5.40	5.40
5.04	RCC Side wall of Rigid Flooring M-30			2.00	5.10	0.20		1.00	2.04	2.04
5.04	RCC concrete block		m ³	36.00	0.30	0.20		0.30	0.65	0.65
		Sub Total								8.09
5.04	RCC in Front Curtain wall		m ³	1.00	6.40		0.81		5.18	5.18
5.04	RCC in Back Curtain wall		m ³	1.00	4.40	0.30		1.00	1.32	1.32
		Sub Total								6.50
5.02	PCC M-15 below RCC Rigid flooring	0.1m thick	m ³	1.00	5.00	5.40		0.10	2.70	2.70
5.02	PCC M-15 below front Curtain wall		m ³	1.00	6.40	1.70		0.10	1.09	1.09
5.20	Hand woven wire crates in flexible apron for protection work	1m thick	m ³	1.00	6.00	6.40		1.00	38.40	38.40
5.07	Reinforcement in RCC flooring		Mt	1.00	8.09		@50kg/m3 of concrete		404.38	0.404
5.07	Reinforcement in curtain wall		Mt	1.00	6.50		@50kg/m3 of concrete		325.20	0.325

Summary Of Quantities for bridges of PP road

Item No. as per Quantity Estimates	Item No. as per BOQ	Description	Unit	RCC Slab at KM 27.165	RCC voided Slab at KM. 31.531	Total quantity	Item No. in Rate Analysis	Remarks
1	2	3	4	5	6	7	8	9
		BILL NO. 6: NEW BRIDGES						
6.01	6.01	Excavation for Foundation of Structure, as per drawings and Technical Specification, in						
	a	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	1075	712	1787	(3.32+3.33)/2	
	b	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum			0	3.34	
	c	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum			0	3.9	
	d	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	461	305	766	3.8 A	
6.02	6.02	PCC (M 15 Grade) in Levelling course below Foundation, Pile Cap, Approach slab etc. complete as per drawings and Technical Specification Section 1700, 2100, 2500 and 2700.	cum	55	48	103	14.10	
6.03	6.03	RCC / PCC in open foundations excluding reinforcement complete as per drawings	cum			0		
	a	M 30 Grade	cum			0	12.8 G case - I	RCC rates are taken
	b	M 35 Grade	cum	353	206	559	12.8 H case - I	RCC rates are taken
6.04	6.04	RCC / PCC in Substructure, Wing Wall, retaining wall, Box Cell Structures, Seismic restrainer etc. excluding reinforcement, in following grades, complete as per Drawings and Technical Specification Section 1700 and 2200				0		
	a	M 25	cum			0	13.5 F (p) Case-I	
	b	M 30	cum			0	13.5 G (p) Case-I	
	c	M 35	cum	382	250	632	13.5 F (h) Case-I	
6.05	6.05	RCC in Superstructure excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2300.				0		
	a	Solid Slab (M 30 Grade)	cum	80	98	177	14.1C Case I (i) q	
	b	Deck Slab for Composite Girder (M35 Grade)	cum			0	14.1D Case I (ii) q	
	c	Voided Slab (M35 Grade)	cum			0	14.1D Case I (i) q	
6.06	6.06	Steel Reinforcement Fe 500D in Foundation, Substructures Superstructure etc. complete as per drawings and Technical Specification Section 1600 .						
	6.07	Steel Girder for Steel Composite Superstructure including railing and fixing of girder with Bearing complete as per drawings and Technical Specification 1000 and 1900.	MT	104	74	178	(12.40+13.6+14.2)/3	Average of three
6.08	6.08	Bituminous (Type 2) Wearing Coat as per drawings and Technical Specification Section 2700.	MT			0	14.25(i)	
6.09	6.09	40 thk. PCC (M25) finished with 15 thk plaster '1:3' complete as per drawings and Technical Specification.	sqm	128	145	272	((5.8*0.040) for grading)+5.14)	
6.10	6.10	Bearings, of following Type, as per drawings and Technical Specification Section 2000	cum	2	2	5	13.5 F (p) Case-I	
	a	Tar Paper Bearings	sqm			0		
	b	Elastomeric Bearings	cum	39	39	77	As per Market	
	c	POT cum PTEE Bearings	cum			0	13.14	

CALCULATION OF QUANTITY OF CRASH BARRIER AND RAILING

Total length of crash barrier = 30 m
Grade of concrete = M40
Unit weight of concrete (t/m^3) = 2.5



Volume of triangle 1 = $0.110 \text{ m}^3/\text{m}$
Volume of triangle 2 = $0.084 \text{ m}^3/\text{m}$
Volume of triangle 3 = $0.045 \text{ m}^3/\text{m}$

Total volume per m length of crash barrier = 0.239 m^3
Dead weight due to per m length of crash barrier = 0.598 t
Total volume of concrete in crash barrier = 7.2 m^3
Reinforcement quantity = 0.7 T
Total length of MS pipe Railing = 30 m
Length of Metal Beam Crash Barrier over Retaining Wall = 20 m
Hand Railing at the edge of Footpath = 30 m

CALCULATION OF QUANTITY FOR RETAINING WALL

Total length of Retaining Wall =	20	m
Overall height of retaining wall =	6	m
Width of Heel Slab =	3.2	m
Width of Toe Slab =	0.6	m
Thickness of Stem at bottom =	0.75	m
Total width of foundation =	4.55	m
Thickness of foundation near Stem =	0.7	m
Thickness of Heel slab Tip =	0.3	m
Thickness of Toe slab Tip =	0.3	m
Thickness of Stem at top =	0.3	m
Height of Stem =	5.3	m
Thickness of filter media behind retaining wall =	0.6	m
Thickness of levelling course below foundation =	0.15	m
Angle of internal friction for Backfill Material =	32	deg
Average height of Ground Level =	3	m
C/s area of Stem =	2.7825	m ²
C/s area of Foundation =	2.425	m ²
C/s area of filter media behind retaining wall =	3.18	m ²
C/s area of backfilling material=	24.745	m ²
C/s area of Earthwork in Excavation =	14.550	
Volume of concrete in Retaining Wall Stem =	55.7	m ³
Volume of concrete in Retaining Wall Foundation =	48.5	m ³
Total Volume of Concrete in Retaining Wall =	104.2	m ³
Volume of filter media behind retaining wall =	64	m ³
Volume of backfilling material behind wall =	495	m ³
Earth work in filling in front of abutment	36	m ³
Volume of levelling course below foundation =	14.3	m ³
Volume of Earthwork in Excavation =	291.0	m ³
Total number of weep holes =	120	Nos
Total HYSD bar for retaining wall @110=	11.5	Mton
Boulder pitching in front of retaining wall =	12	m ³

SUMMARY FOR QUANTITY CALCULATION FOR BRIDGE AT KM - 31.84

Item No.	Description	Unit	Value
6.01	for soft rock	cum	712.04
	for hard rock with no blast	cum	305.16
6.02	PCC Levelling Course (M15)	cum	48.15
6.03	RCC/PCC in open foundation	cum	205.57
6.04	RCC/PCC in Substructure	cum	249.73
6.05	RCC in Superstructure	cum	97.83
6.06	Steel Reinforcement	MT	74.24
	HYSD bar reinforcement complete as per drawings and technical specifications:		
	Abutment =	MTon	55.4
	Pier =	MTon	0.0
	Retaining wall =	MTon	8.3
	Crash Barrier @100	MTon	0.8
	Superstructure @100 =	MTon	9.8
	Total =	MTon	74.2
6.08	Bituminous Wearing coat	sqm	144.50
6.09	Wearing coat on Footpath	sqm	2.43
6.10	Tar paper bearing	sqm	38.64
6.11	Expansion joints	m	26.00
6.13	Drainage Spouts	Nos	13.00
6.14	Approach Slab	cum	33.08
6.15	Length of hand rails	Rm	34.00
6.16	Crash Barrier	cum	8.14
6.17	Earth work in Filling	cum	0.00
6.18	Weep holes	Nos	245
6.19	Backfilling behind earth retaining structures	cum	1240
6.20	Filter media behind earth retaining structures	cum	153
6.21	Boulder grouted with cement mortar	cum	121.80
6.22	PCC M-15 in annular space	cum	59.98
6.23	Preparation of rock foundation	sqm	202.50
6.24	Lettering and No. of bridges	Nos	40.00
6.25	Coats of cement paint on crash barrier and railing	sqm	340.00

Annexure -8.01

Chainage		Length	Type of Cross Section
From	To		
0+000	0+655	655	IV
0+655	1+125	470	I
1+125	1+225	100	II
1+225	2+315	1090	I
2+315	2+360	45	III
2+360	5+320	2960	I
5+320	5+440	120	III
5+440	7+958	2518	I
7+958	8+088	130	III
8+088	11+436	3348	I
11+436	11+467	31	III
11+467	15+800	4333	I
15+800	16+370	570	IV
16+370	19+140	2770	I
19+140	19+160	20	III
19+160	22+167	3007	I
22+167	22+206	39	III
22+206	27+213	5007	I
27+213	27+230	17	III
27+230	30+815	3585	I
30+815	30+878	63	III
30+878	31+545	667	I
31+545	31+619	74	III
31+619	33+500	1881	I
33+500	33+530	30	III
33+530	34+055	525	I
34+055	34+084	29	III
34+084	34+355	271	I
34+355	34+410	55	III
34+410	40+864	6454	I
40+864	40+904	40	III
40+904	43+975	3071	I
43+975	44+005	30	III
44+005	46+400	2395	I
46+400	46+435	35	III
46+435	47+752	1317	I
47+752	47+783	31	III
47+783	49+670	1887	I
49+670	49+760	90	III
49+760	51+496	1736	I
51+496	51+510	14	III
51+510	62+250	10740	I
62+250	62+558	308	IV

Length of Typical Cross Section	
Type	Length
I	60032
II	100
III	893
IV	1533
V	
Bridge	
Total	62558

Calculation For Foundation Excavation Quantity and Backfill Quantity for Different Height of Gabion Wall/ Unit Run

Gabion Ht Upto GL =	4.0 m	4.5 m	5.0 m	5.5 m	6.0 m	6.5 m	7.0 m
Base Width + Projection =	3.5 m	4.0 m	4.0 m	4.5 m	4.5 m	5.0 m	5.0 m
Foundation Depth at Valley end =	1.5 m	1.5 m	1.5 m	1.5 m	1.5 m	1.5 m	1.5 m
Hill Slope angle =	50.00	50.00	50.00	50.00	50.00	50.00	50.00
Cut Slope Angle =	60.00	60.00	60.00	60.00	60.00	60.00	60.00
Height of metal thickness =	0.500	0.500	0.500	0.500	0.500	0.500	0.500
Height of cutting area above valley end (H)=	16.680	18.590	18.590	20.501	20.501	22.411	22.411
Height of Gabion wall=	5.5	6.0	6.5	7.0	7.5	8.0	8.5
Height of Gabion wall above valley end =	4.0	4.5	5.0	5.5	6.0	6.5	7.0
Exca.Area for GW Foundation =	5.90	6.65	6.65	7.40	7.40	8.15	8.15
Exca. Area above Gabion Wall Foundation =	36.41	45.23	45.23	55.00	55.00	65.73	65.73
Base for Extra Counted Area=	3.45	3.82	3.69	4.06	3.93	4.30	4.16
Height for Extra Counted Area =	13.18	14.59	14.09	15.50	15.00	16.41	15.91
Extra counted area (sqm)=	22.74	27.86	25.98	31.45	29.45	35.25	33.13
Total Excavated area (sqm)=	19.58	24.02	25.90	30.96	32.95	38.63	40.75
Cross Section Area (sqm) =	13.25	16.75	17.25	21.25	21.75	26.25	26.75
Back Filling Area 1(sqm) =	8.73	10.39	12.20	14.15	16.24	18.48	20.86
Back Filling Area 2 (sqm) =	0.75	1.25	2.75	3.5	5.25	6.25	8.25
Total Back Filling area (sqm)=	9.48	11.64	14.95	17.65	21.49	24.73	29.11

Station/ Chainage (m)	Type of Retaining Structure	Retain HL	Retain Length	Culvert Position	X-Area of GW (m ²)	Surface Area of RE Wall (m ²)	Volume of GW (m ³)	RE Wall Actually Provided GW (m ²)	Add Qty GW in place replaced RE Wall (m ²)	Deducted Qty GW at Culvert location (m ²)	Exc. Qty for GW (m ³)	Backfill Qty for GW (m ³)	Exc. Qty for REW (m ³)	Backfill Qty for REW (m ³)
9320	GW	2.5	1.37		9.25	0.00	12.66				16.92	6.32	0.00	0.00
9330	GW	3.5	10.00		12.75	0.00	127.50				178.20	72.17	0.00	0.00
9340	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
9350	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
9360	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9360.631	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9370	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9380	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9390	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9391.791	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9400	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9405.291	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9410	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9420	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9420.291	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9430	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9440	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9450	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9460	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9470	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9480	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9490	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00
9500	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00
9510	GW	3.0	10.00		9.75	0.00	97.50				139.97	58.46	0.00	0.00
9520	GW	3.0	10.00		9.75	0.00	97.50				139.97	58.46	0.00	0.00
9525.675	GW	3.0	5.08		9.75	0.00	49.48				71.03	29.67	0.00	0.00
9530	GW	2.5	4.52		9.25	0.00	45.56				60.86	22.75	0.00	0.00
9540	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
9546.504	GW	2.0	6.50		4.63	0.00	39.08				49.78	23.00	0.00	0.00
9550	GW	2.0	3.50		4.63	0.00	16.17				26.76	12.36	0.00	0.00
9560	GW	2.0	10.00		4.63	0.00	46.25				76.54	35.36	0.00	0.00
9565.719	GW	2.0	5.79		4.63	0.00	26.77				44.31	20.47	0.00	0.00
9570	GW	2.0	4.21		4.63	0.00	19.48				32.23	14.89	0.00	0.00
9580	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9590	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9600	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9610	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9620	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9630	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9640	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9644.256	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9644.356	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9644.357	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9650	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9660	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9670	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9680	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9690	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9700	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9710	GW	1.5	10.00	C	4.13	0.00	41.30			33.04	63.85	25.98	0.00	0.00
9720	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9722.56	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9730	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9740	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9750	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9760	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9770	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9780	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9785.48	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9790	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9800	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9800.48	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9810	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9813.96	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9820	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9830	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9840	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9850	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9860	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9870	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9880	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9890	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9900	GW	2.0	10.00		4.63	0.00	46.25				76.54	35.36	0.00	0.00
9905.892	GW	2.0	5.89		4.63	0.00	27.25				45.10	20.84	0.00	0.00
9910	GW	3.0	4.11		9.75	0.00	40.05				57.50	24.01	0.00	0.00
9920	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
9925.178	GW	2.5	5.18		9.25	0.00	47.90				63.99	23.92	0.00	0.00
9930	GW	2.0	4.82		4.63	0.00	22.30				36.91	17.05	0.00	0.00
9940	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9946.607	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9950	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9960	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9970	Para W	0.0	0.00	C	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
9980	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9990	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10000	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10010	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10020	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10029.189	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10030	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10040	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10044.183	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10050	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10050.016	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10060	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10070	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10080	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10083.212	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10083.312	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10083.313	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
10090	Para W	0.0	0.00		0.00									

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in State of Nagaland under SARDP-NE

RTES Ltd.
Highway Division

Quantity Calculation for Traffic Signs

Annexure -9.01

SL.NO.	NAME	CODE	LOCATION OF CAUTIONARY SIGNS															TOTAL
			KM53-KM54	KM54-KM55	KM55-KM56	KM56-KM57	KM57-KM58	KM58-KM59	KM59-KM60	KM60-KM61	KM61-KM62	KM62-KM62.558						
	STOP	14.01															37	
	GIVE WAY	14.02										4					3	
	GIVE WAY TO BUSES EXISTING THE BUS BAY	14.03															0	
	NO STOPPING OR STANDING	14.28															0	
	NO PARKING	14.29															0	
	MAXIMUM SPEED LIMIT	14.37										1					1	
	COMPULSORY KEEP LEFT	14.48															3	
	COMPULSORY KEEP RIGHT	14.49															0	
	COMPULSORY SOUND HORN	14.57															0	
	LEFT HAND CURVE	15.01		4													2	
	RIGHT HAND CURVE	15.02					1										82	
	RIGHT HAIRPIN BEND	15.03															16	
	LEFT HAIRPIN BEND	15.04															0	
	RIGHT HAIRPIN BEND	15.04A															0	
	LEFT REVERSE BEND	15.05		1	2	2	1			2	1	2	1				63	
	RIGHT REVERSE BEND	15.06		1	2	2	1			2	1	2	1				63	
	SERIES OF BEND	15.07		4	2	2	2			2	2	2	2	2			120	
	SIDE ROAD RIGHT	15.09															41	
	SIDE ROAD LEFT	15.10							2								41	
	Y-INTERSECTION LEFT	15.11											3				0	
	Y-INTERSECTION RIGHT	15.12															0	
	Y-INTERSECTION	15.13															0	
	X-ROAD	15.14															0	
	ROUNDBOUT	15.15															0	
	T-INTERSECTION	15.17															0	
	T-INT. MAJOR ROAD	15.18															0	
	STAGGERED INTERSECTION RIGHT	15.20															0	
	STAGGERED INTERSECTION LEFT	15.21															0	
	NARROW BRIDGE AHEAD	15.25															0	
	STEEP ASCENT	15.26		1		1	1			1	1	2	1	1			18	
	STEEP DESCENT	15.27		1		1	1			1	1	2	1	1			18	
	PEDESTRIAN CROSSING	15.33										2	2	2			32	
	SCHOOL AHEAD	15.34															2	
	BUILT UP AREA	15.35										2	2	2			28	
	PLAYGROUND AHEAD	15.58										2					2	
	BARRIER	15.60															0	
	FALLING ROCKS	15.65															0	
	SINGLE CHEVRON (SMALL)	15.72		33	35	38	35	34	37	30	33	38	29				2290	
	DOUBLE CHEVRON	15.74															0	
	OBJECT MARKER(LEFT)	15.76		8	9	6	6	10	6	8	5	9	3				439	
	OBJECT MARKER(RIGHT)	15.77		8	9	6	6	10	6	8	5	9	3				439	
	TWO WAY HAZARD MARKER	15.78															0	
	MAP TYPE ADVANCE DIRECTION SIGN	16.02								3			3				9	
	REASSURANCE SIGN	16.05															0	
	FLAG TYPE DIRECTION SIGN	16.04										2	2				52	
	PLACE IDENTIFICATION SIGN	16.06									2	2	2				20	
	TRUCK LAY- BY	16.07															0	
	WEIGH BRIDGE AHEAD	16.09															0	
	GANTORY BOARD	16.10															0	
	TOILET	17.05														1	2	
	FILLING STATION	17.06															0	
	HOSPITAL	17.07															0	
	POLICE STATION	17.13															0	
	PICNIC SITE	17.14															0	
	HOME ZONE	17.21															0	
	BUSTOP	17.35									2		2				2	
	STATE HIGHWAY ROUTE MARKER SIGN	20.01															18	
	NATIONAL HIGHWAY ROUTE MARKER SIGN	20.02								5							0	
	TOTAL NO OF SIGNS		61	59	60	55	58	59	78		65	84	75				3875	

Quantity Calculation for Traffic Signs

SL.NO.	NAME	CODE	LOCATION OF CAUTIONARY SIGNS																		
			KM36-KM37	KM37-KM38	KM38-KM39	KM39-KM40	KM40-KM41	KM41-KM42	KM42-KM43	KM43-KM44	KM44-KM45	KM45-KM46	KM46-KM47	KM47-KM48	KM48-KM49	KM49-KM50	KM50-KM51	KM51-KM52	KM52-KM53		
	STOP	14.01							1												
	GIVE WAY	14.02																			
	GIVE WAY TO BUSES EXISTING THE BUS BAY	14.03																			
	NO STOPPING OR STANDING	14.28																			
	NO PARKING	14.29																			
	MAXIMUM SPEED LIMIT	14.37																			
	COMPULSORY KEEP LEFT	14.48																			
	COMPULSORY KEEP RIGHT	14.49																			
	COMPULSORY SOUND HORN	14.57	2	2				2	2	2					2		2				
	LEFT HAND CURVE	15.01						2	1												
	RIGHT HAND CURVE	15.02						2													
	RIGHT HAIRPIN BEND	15.03						2	1												
	LEFT HAIRPIN BEND	15.04																			
	RIGHT HAIRPIN BEND	15.04A																			
	RIGHT REVERSE BEND	15.05	1	1	1					1											
	LEFT REVERSE BEND	15.06	1	1	1					1				1			1		1		
	SERIES OF BEND	15.07	2	2	2			2		2	4			1	2	2	2		1		
	SIDE ROAD RIGHT	15.09			2			2						4	2	2	2		1		
	SIDE ROAD LEFT	15.10			2					1					1	2	2		1		
	Y-INTERSECTION LEFT	15.11			2					1					1	2			2		
	Y-INTERSECTION RIGHT	15.12																			
	Y-INTERSECTION	15.13																			
	X-ROAD	15.14																			
	ROUNDABOUT	15.15																			
	T-INTERSECTION	15.17																			
	T-INT. MAJOR ROAD	15.18																			
	STAGGERED INTERSECTION RIGHT	15.20																			
	STAGGERED INTERSECTION LEFT	15.21																			
	NARROW BRIDGE AHEAD	15.25																			
	STEEP ASCENT	15.26	1		1					1											
	STEEP DESCENT	15.27	1		1					1											
	PEDESTRIAN CROSSING	15.33			2																
	SCHOOL AHEAD	15.34			2																
	BUILT UP AREA	15.35																			
	PLAYGROUND AHEAD	15.58		1	1									1	1						
	BARRIER	15.60																			
	FALLING ROCKS	15.65																			
	SINGLE CHEVRON (SMALL)	15.72	30	29	29		32	38	41	36	41	39	43	43	29	46	39	40	35		
	DOUBLE CHEVRON	15.74																			
	OBJECT MARKER(LEFT)	15.76	4	5	7		10	9	9	8	14	12	10	9	10	8	7	8	10		
	OBJECT MARKER(RIGHT)	15.77	4	5	7		10	9	9	8	14	12	10	9	10	8	7	8	10		
	TWO WAY HAZARD MARKER	15.78																			
	MAP TYPE ADVANCE DIRECTION SIGN	16.02																			
	REASSURANCE SIGN	16.05																			
	FLAG TYPE DIRECTION SIGN	16.04		2	4											2					
	PLACE IDENTIFICATION SIGN	16.06																			
	TRUCK LAY- BY	16.07		1	1										1	1					
	WEIGH BRIDGE AHEAD	16.09																			
	GANTORY BOARD	16.10																			
	TOILET	17.05																			
	FILLING STATION	17.06																			
	HOSPITAL	17.07																			
	POLICE STATION	17.13																			
	PICNIC SITE	17.14																			
	HOME ZONE	17.21																			
	BUSTOP	17.35																			
	STATE HIGHWAY ROUTE MARKER SIGN	20.01			2											2					
	NATIONAL HIGHWAY ROUTE MARKER SIGN	20.02																			
	TOTAL NO OF SIGNS		46	57	63	54	58	67	62	59	75	68	67	69	59	84	57	62	61		

Calculation of Quantity for Overhead Gantry

S. No.	Description	Unit	No	Length (m)	Breadth (m)	Height / Depth (m)	Unit weight (Kg/m)	Quantity (Kg)
(A) Truss								
	80mm dia NB Pipe	m	4	14.10			8.36	471.50
	40mm dia NB Pipe at Side Elevation							
	Vertical Side Pipe	m	28	1.24			3.56	123.60
	Horizontal Pipe	m	4	0.74			3.56	10.54
	Diagonal pipe at ends	m	4	1.56			3.56	22.21
	Diagonal pipe at midway	m	22	1.63			3.56	127.66
	25mm dia NB Pipe at top & bottom Elevation							
	Diagonal pipe at ends	m	4	1.22			2.41	11.76
	Diagonal pipe at midway	m	22	1.30			2.41	60.93
	Cross Pipe	m	24	0.67			2.41	38.75
(B) Vertical Support of Pipe 300 mm Dia								
		m	2	5.00			49.30	493.00
Total wt (A+B)								1367.96
S. No.	Description	Unit	No	Length (m)	Breadth (m)	Height / Depth (m)	Unit weight (Kg/cum)	Quantity
(C) Gusset Plates								
	Gusset Plate at top of Truss Frame	sqm	4	1.00	1.00	0.01	7854	376.99
	Stiffners for Gusset Plate at Top frame of Truss	Sqm	8	0.56	0.30	0.01	7854	105.56
	Frame	sqm	4	1.00	1.00	0.01	7854	376.99
	Stiffners for Gusset Plate at Base of Truss Frame	Sqm	8	0.56	0.30	0.01	7854	105.56
	M.S. Plates welded to Bolts	Sqm	16	0.10	0.10	0.01	7854	12.57
Total wt (C)								977.67
Total weight of One Overhead Gantry (A+B+C) in (kg)								2345.63

Location of Proposed Gantry Signs

S. No.	Location	No. of Gantry Signs
1	Start of Project Road	1
2	End of Project Road	1
Total		2

S. No.	Description	Unit	No. of Overhead Gantry	No	Length (m)	Breadth (m)	Height / Depth (m)	Quantity	Say
1	Gantry Mounted Overhead Signs	Sqm	2	6	2.30	1.30		35.88	40
2	Foundation Pedestal								
	Excavation	cum	2	2	2.40	2.40	2.10	48.38	50
	PCC M-15 in footing	cum	2	2	2.40	2.40	0.10	2.30	3
	M-25 RCC in Footing	cum	2	2	2.20	2.20	0.30	5.81	6
	M-25 RCC in Pedestal	cum	2	2	1.00	1.00	2.45	9.80	10
	Reinforcemnt @ 60kg/cum of concrete	tonne	2					1.87	2
3	Overhead Gantry	tonne	2					4.69	5

Quantity Calculation for Roadway Indicators

No. of Curves	Length of Curve (m)	Radius of Curve (m)	Spacing of Roadway Indicators (m)	Nos. of Roadway Indicators
0	0	<21	6	0
0	0.00	21 -33	6	0
80	5545.88	33	6	1004
47	3384.05	33-41	8	470
117	8559.12	41-61	8	1187
169	12594.51	61-101	12	1219
192	16625.63	101-301	20	1023
26	2758.69	>301	25	136
Total No. of Roadway Indicators				5039

Calculation of Quantities for Junction & Median Improvement

S No.	Chainage (Km)	Type of Junction	Island in Sq mt	Area Road in Sq mt	E/S Area (Cross Leg) in Sq mt	Length of Kerb on Island and Median (Lm)	No. of Arrows		Type of Junction
							Straight	Straight-Left/Right	
1	0/000	Major	525	4425	620	547	12	4	T-Junction
2	59/030	Major	250	2175	435	270	8	0	Y-Junction
3	62/550	Major	274	2976	500	353	8	3	X-Junction
		Minor		4250	2250				
Total			1049.00	13826.00	3805.00	1170.00	28.00	7.00	

SUMMARY OF QUANTITIES

S No.	Description	unit	No	Area (sqm)	Th. (m)	Qty.	Say
1	BC	cum	1	13826.00	0.030	414.78	415.00
2	DBM	cum	1	13826.00	0.050	691.30	691.00
3	WMM	cum	1	13826.00	0.250	3456.50	3457.00
4	GSB	cum	1	13826.00	0.210	2903.46	2903.00
5	Subgrade	cum	1	17631.00	0.500	8815.50	8816.00
6	Prime Coat	sqm	1	13826.00		13826.00	13826.00
7	Tack Coat	sqm	1	13826.00		13826.00	13826.00
8	Kerb Stone in Island & Medians	m		1170.00		1170.00	1170.00
9	Soil Filling in Medians & Island	cum	1	1049.00	0.710	744.79	745.00

GSB = $4425 \times 0.21 = 929.25$

WMM = $4425 \times 0.25 = 1106.25$

DBM = $4425 \times 0.05 = 221.25$

BC = $4425 \times 0.03 = 132.75$

PC = $4425 = 4425$

TC = $4425 = 4425$

Kerb Stone = 547 m

Soil filling in median & island = $525 \times 0.710 = 372.75$